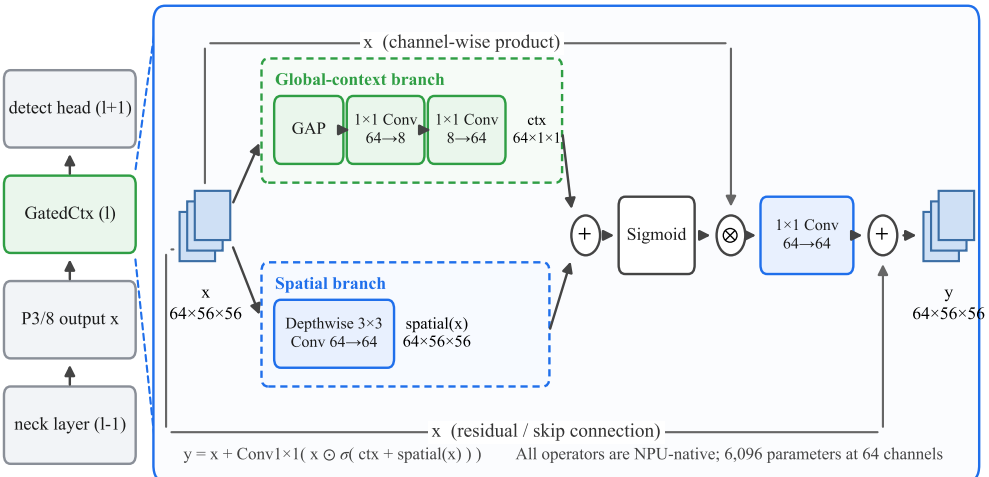
detect head (l+1)
GatedCtx (l)
P3/8 output x
neck layer (l-1)
x (channel-wise product)
Global-context branch
GAP
1×1 Conv 64→8
1×1 Conv 8→64
ctx 64×1×1
x 64×56×56
Spatial branch
Depthwise 3×3 Conv 64→64
spatial(x) 64×56×56
Sigmoid
1×1 Conv 64→64
y 64×56×56
x (residual / skip connection)
y = x + Conv1×1(x ⊙ σ(ctx + spatial(x)))
All operators are NPU-native; 6,096 parameters at 64 channels